

Computer-Use Automation System

Design report | Public repository preparation | 13 September 2026

Architecture

This system discovers and replays a savings-balance lookup against a local Flask bank containing synthetic members. The workflow is search, member record, accounts, then balance and currency. Python controllers share one executor and one Playwright/Chromium surface. Keeping orchestration in one asyncio process makes browser lifetime and exclusive control explicit without introducing queues or services that the demonstration does not need.

Discovery accepts a goal and reviewed target configuration. Gemini or OpenAI returns structured actions; validation and policy run before execution. Gemini 2.5 Flash supplied six actual decisions in the preserved discovery run. The model sees reviewed control descriptions and live visibility/state flags, with input references instead of member values. It chooses the sequence within an approved vocabulary; it does not discover arbitrary locators on unknown websites. This limits generality while keeping model output and application text untrusted.

Artifact schema

Pydantic contracts describe schema version, capability identity/version, compatible target, typed inputs/outputs, ordered steps, locator rationale, parameter references, success checkpoint, business outcomes, and bounded recovery rules. The balance is a validated decimal string, avoiding binary floating-point loss. Unknown actions, incompatible versions, missing inputs, bad references, and undeclared outputs are rejected before execution.

Compilation includes only successful actions and restores input references from known policy templates, never by replacing every matching string. The artifact is independent of the transcript and has explicit provenance and a discovery run ID. Genuine artifacts and hand-authored fixtures remain distinct. Immutable files and SHA-256 manifests support review and correlation; provenance is not a provider-signed attestation. Replay uses the same artifact for a different member and reads that member's current UI values.

Determinism & error handling

Replay binds inputs and follows recorded steps without model decisions. Exact scoped roles/labels are preferred; reviewed CSS identifies legacy-style balance cells. Missing or ambiguous controls fail rather than selecting the first match. Condition-based waits share step deadlines. Success requires typed outputs and a visible account-owner checkpoint; completed clicks alone are insufficient.

The result separates success, member-not-found, and hard failures such as denied permission, session expiry, app errors, invalid input, and failed checkpoints. Known loading permits bounded waits, with no repeated search submission. Exhausted recovery stops and requests intervention. Discovery additionally limits steps, elapsed time, invalid responses and repeated states. Model declarations cannot supply the result. Persistence failure closes the browser instead of reporting an unaudited success.

Heterogeneity & multi-tenant

Controllers use a surface contract rather than Playwright objects. A desktop adapter could resolve OS accessibility controls, with reviewed screenshot or coordinate targeting where accessibility is absent. Legacy frames require explicit frame scopes and separately tested navigation boundaries. Neither adapter is implemented; silently guessing visual targets would weaken safety.

Tenant reuse would retain the capability while applying reviewed origin, identity, locator and operation configuration for a compatible app version. Each run already has an isolated browser context. Product/version markers and exact route checks reject incompatible targets; markers assume a cooperating app and are not authentication. New tenants need contract tests and policy review before reuse. A deployed tenant registry, credential vault and cross-tenant compatibility matrix are future infrastructure, not features claimed by this demonstration.

Escalation & handoff

Interactive runs keep the existing browser alive and expose a token-protected loopback operator screen with goal, stopped action, reason and sanitized context. Ownership proceeds through `AUTOMATION_RUNNING`, `AWAITING_HUMAN`, `HUMAN_CONTROL` and `RESUME_CHECK`. Automation dispatch is blocked during human ownership; cookies, page and session identity survive. Trusted browser events record interaction categories and element tags without field contents or keypress data.

Resume verifies target compatibility, the exact requested account route, owner checkpoint and output targets. A single-use permission allows fresh output reads and final verification without repeating navigation the operator completed. Wrong pages remain paused. Cancel, operator deadline, closed windows and interrupted control commands terminate safely. Human waiting has a separate bounded budget. Human-assisted discovery may finish with verified outputs, but cannot compile an incomplete model-only action history into a capability.

A separate manual operator run captured click, submit and navigation events, then resumed successfully in the same session. Automated handoff tests are labelled simulated-operator checks and are not presented as manual evidence.

Safety

Both modes enforce default-deny origins, routes, actions and operations through the shared executor. Actual controls, form methods and destinations are checked; redirect destinations are checked before following. Frames, popups, background APIs, WebSockets and downloads are blocked. A separate human-only rule permits just the current member's CSRF-checked dialog acknowledgement. It grants no transfer operation or general POST permission. Model or UI text cannot widen it.

Diagnostics use per-run keyed tokens for unreviewed text and mask registered input/output values. Structural failure snapshots retain hierarchy and visibility, not text, values, URLs, cookies or scripts. No raw screenshots or traces are saved. Keys remain outside Git. The local operator token protects commands but does not establish a person's identity. Browser locks gate automation; cooperative DOM suppression cannot lock the OS, address bar or developer tools. This is a bounded local demonstration, not a production banking security boundary.

Cuts

The implemented vertical slice includes genuine model discovery, generated capability reuse, deterministic outcomes, policy, redacted evidence and live human handoff. Fresh-source acceptance installs a new locked environment, runs the full suite, applies replay/handoff checks to the genuine artifact, and captures eight CLI scenarios. Historical discovery and manual handoff records are preserved; acceptance reports retain failed attempts rather than rewriting them as successes. The reviewer commands and evidence map are in docs/acceptance.md.

The current source snapshot identifies the accepted runtime; no exact source snapshot was captured for the older discovery run. Caches and the same macOS host are reused, so this is not independent OS validation. Native desktop execution, unknown-site exploration, multi-user operation, production credential management and broad workflow repair are deliberate cuts. Next work would prioritize a second target adapter, reviewed tenant variants and a production threat model. Public GitHub publication is authorized after final review. Email submission remains pending explicit user direction.